Chronic Kidney Disease (CKD) and Its Complications

All key points for the exam

1. Definition (KDIGO)

Abnormal kidney structure or function present for **more than 3 months**, shown by **either** of the following:

Criterion	Value
Decreased GFR	**GFR < 60 mL/min/1.73 m²**
Or markers of kidney damage (one or more)	**Albuminuria** (**ACR ≥ 30 mg/g**) • Abnormal urine sediment (e.g., RBC casts) • Electrolyte abnormalities from tubular disorders • Abnormal histology (biopsy) • Abnormal imaging (e.g., polycystic kidneys) • **History of kidney transplant**

Duration > 3 months is what separates **CKD** from **AKI**.

2. Staging (GFR + Albuminuria)

GFR categories

Stage	GFR (mL/min/1.73 m²)	Description
G1	**≥ 90**	Normal or high (CKD only if there are markers of damage)
G2	**60–89**	Mildly decreased
G3a	**45–59**	Mildly to moderately decreased
G3b	**30–44**	Moderately to severely decreased
G4	**15–29**	Severely decreased
G5	**< 15**	**Kidney failure** (ESKD if on dialysis)

Albuminuria categories

Category	ACR (mg/g)	Description
A1	**< 30**	Normal to mildly increased
A2	**30–300**	Moderately increased (formerly "microalbuminuria")
A3	**> 300**	Severely increased (formerly "macroalbuminuria")

Example: "CKD **G3b A2**" means GFR 30–44 with ACR 30–300.
ACR = urine albumin-to-creatinine ratio (spot urine sample; **preferred** test).
Higher G stage + higher A stage = **higher risk** of progression and cardiovascular events.

3. Causes and Risk Factors

Cause	Note
Diabetes	**#1 cause**
Hypertension	**#2 cause**
Glomerulonephritis	
Polycystic kidney disease (PKD)	Tolvaptan for rapidly progressive ADPKD

Cause	Note
Others	Recurrent AKI, nephrotoxic drugs, obstruction, autoimmune disease (lupus)

Factors that speed progression: uncontrolled BP, **uncontrolled diabetes**, **proteinuria/albuminuria**, **smoking**, **obesity**, **dyslipidemia**, **nephrotoxins** (especially NSAIDs), **metabolic acidosis**, **recurrent AKI**.

4. Estimating Kidney Function

Method	Use
CKD-EPI 2021 equation (race-free)	**Diagnosis and staging** of CKD (preferred)
CKD-EPI creatinine + cystatin C	More accurate when creatinine is unreliable (extremes of muscle mass, amputees, malnutrition)
Cockcroft-Gault (CrCl)	**Drug dosing** (most drug labels are based on it)

Cockcroft-Gault: $CrCl = [(140 - age) \times weight\ (kg)] \div [72 \times SCr]$ → **× 0.85 for females**

Creatinine is affected by **muscle mass**: **elderly, frail, amputees** → creatinine looks "normal" while GFR is actually low.

5. Slowing CKD Progression (most important section)

A) Blood pressure control

Guideline	BP target
KDIGO 2021	**Systolic < 120 mmHg** if tolerated (standardized office measurement)
ACC/AHA 2017	**< 130/80 mmHg**

B) ACE inhibitors or ARBs

Point	Details
Indication	**CKD with albuminuria** (**A2 or A3**), **especially with diabetes or hypertension**
Mechanism of benefit	**Efferent arteriole dilation** → **lower intraglomerular pressure** → **less proteinuria** → slower progression
Dose	**Titrate to the maximum tolerated dose**
Monitoring	**SCr and potassium within 2–4 weeks** after starting or increasing the dose
Creatinine rise	**Up to 30%** is **acceptable** → continue • **> 30%** → reduce or stop, look for renal artery stenosis or volume depletion
Hyperkalemia	Manage potassium (diet, diuretics, binders) **before stopping** the ACEi/ARB
Do not combine	**ACEi + ARB**, or either with **aliskiren** → more hyperkalemia and AKI, no benefit
Pregnancy	**Contraindicated** (teratogenic)

C) SGLT2 inhibitors (now a cornerstone)

Point	Details
Drugs	**Dapagliflozin**, **empagliflozin** (CKD with or without diabetes), **canagliflozin** (CKD with type 2 diabetes)
Indication	**CKD with eGFR ≥ 20**, **with or without diabetes** (especially with albuminuria or heart failure)
Continue	**Until dialysis or transplant**, even if eGFR falls below 20
Initial eGFR dip	**Expected and acceptable** (hemodynamic effect); **do not stop** the drug
Benefit	**Kidney and cardiovascular protection** independent of glucose lowering (glucose-lowering effect is weak at low eGFR)
Side effects	**Genital mycotic infections**, UTIs, **volume depletion**, **euglycemic DKA**, (canagliflozin: amputations, fractures)
Sick day rule	**Hold** during acute illness, dehydration, and **3–4 days before surgery**

D) Finerenone (nonsteroidal mineralocorticoid receptor antagonist)

Point	Details
Indication	**Type 2 diabetes + CKD with albuminuria (ACR ≥ 30)**, on max tolerated ACEi/ARB
Start if	**eGFR ≥ 25** and **K ≤ 5.0 mEq/L**
Main risk	**Hyperkalemia** → monitor potassium
Interaction	**CYP3A4 substrate**: avoid strong CYP3A4 inhibitors

E) GLP-1 receptor agonists

- **Semaglutide** (**FLOW trial**) **reduces kidney outcomes** in **type 2 diabetes + CKD**.
- Preferred glucose-lowering agent if more glucose control is needed, or if SGLT2i is not tolerated.

F) Diabetes control

- **A1c target ~ < 7%**, individualized (less strict in advanced CKD because of **hypoglycemia** risk).
- **Insulin and sulfonylureas** are **cleared less** in CKD → **higher hypoglycemia risk** → reduce doses.

Metformin dosing by eGFR

eGFR	Action
≥ 45	Full dose
30–44	**Do not start**; if already on it, **reduce dose** (max **1,000 mg/day**)
< 30	**Contraindicated** (lactic acidosis)

G) Lipids

- **Statin** (or statin + ezetimibe) for **all CKD patients ≥ 50 years** not on dialysis.
- **Dialysis patients: do not start** a statin, but **continue** it if already taking one.

H) Lifestyle

Intervention	Target
Protein intake	**0.8 g/kg/day** in CKD G3–G5 **not on dialysis** • **1.0–1.2 g/kg/day** on **dialysis** (protein is lost in dialysis)
Sodium	**< 2 g/day**
Smoking	**Stop**
Weight, exercise	Healthy weight, regular activity
Avoid nephrotoxins	**NSAIDs**, contrast if possible

I) Metabolic acidosis

- **Treat if serum bicarbonate < 18 mEq/L** (KDIGO) (some references use **< 22**) → slows progression and improves bone and muscle health.
- **Sodium bicarbonate** or **sodium citrate**.
- **Citrate increases aluminum absorption → never give citrate with aluminum-containing binders**.

6. Complications of CKD

Most complications start or worsen from **CKD G3** onward.

Complication	Cause
Anemia	↓ **Erythropoietin (EPO)** + iron deficiency
CKD-mineral and bone disorder (CKD-MBD)	↑ Phosphate, ↓ active vitamin D, ↓ calcium, ↑ **PTH**
Metabolic acidosis	↓ Acid excretion
Hyperkalemia	↓ Potassium excretion
Fluid overload / edema	↓ Sodium and water excretion
Hypertension	Volume overload + RAAS activation
Cardiovascular disease	**#1 cause of death** in CKD
Uremia	Accumulation of waste products
Dyslipidemia	
Malnutrition	Anorexia, protein restriction, dialysis losses

7. Anemia of CKD

Cause

- **Main cause: ↓ erythropoietin (EPO)** production by the kidney.
- **Other causes: iron deficiency** (most common correctable cause), blood loss (dialysis, GI), **shortened RBC survival**, inflammation, B12/folate deficiency, hyperparathyroidism.
- **Type:** usually **normocytic, normochromic**.

Workup

- **Start evaluation when Hgb < 13 g/dL (males) or < 12 g/dL (females).**
- Check: **CBC**, **reticulocyte count**, **iron studies** (**TSAT**, **ferritin**), **B12**, **folate**.

Test	Meaning
TSAT (transferrin saturation)	Iron **available** for RBC production
Ferritin	**Iron stores** (**acute phase reactant**: rises with inflammation, so it may be falsely normal or high)

Step 1: Iron therapy (always correct iron first)

> **Give iron if TSAT ≤ 30% and ferritin ≤ 500 ng/mL** (KDIGO).
> **Hold iron if ferritin > 500–800** or **TSAT > 50%** (iron overload), and **during active infection**.

Setting	Preferred route
CKD not on dialysis, peritoneal dialysis	**Oral iron** first (IV if no response or intolerance)
Hemodialysis	**IV iron** (oral iron is poorly absorbed, and IV access is available)

Oral iron

Salt	Elemental iron
Ferrous sulfate 325 mg	**65 mg (20%)**
Ferrous gluconate 325 mg	**~36 mg (12%)**
Ferrous fumarate 324 mg	**~106 mg (33%)**

- **Dose:** about **200 mg elemental iron/day** (or alternate-day dosing for better absorption and tolerance).
- **Take on an empty stomach**; vitamin C may help.
- **Separate from phosphate binders**, **antacids**, **PPIs**, **calcium**, **tetracyclines**, **fluoroquinolones**, **levothyroxine**.
- **Side effects: constipation**, nausea, **dark/black stools**.

IV iron

Product	Key points
Iron dextran (INFeD)	**Boxed warning: anaphylaxis → test dose (25 mg)** required
Iron sucrose (Venofer)	Common in HD (e.g., 100 mg per HD session × 10 = 1 g)
Sodium ferric gluconate (Ferrlecit)	e.g., 125 mg × 8 doses
Ferumoxytol (Feraheme)	510 mg × 2 doses (3–8 days apart) • **Interferes with MRI** • hypersensitivity
Ferric carboxymaltose (Injectafer)	**Hypophosphatemia**
Ferric derisomaltose (Monoferric)	Large single doses

Ferric pyrophosphate citrate (Triferic) is added to the **dialysate** to maintain iron.

Step 2: Erythropoiesis-stimulating agents (ESAs)

Point	Details
Start ESA	When **Hgb < 10 g/dL**, **after iron is replete** and other causes are ruled out
Goal	**Avoid transfusion**; **do not normalize Hgb**
Target	**Do not exceed Hgb 11.5 g/dL** (KDIGO). FDA label: **reduce or interrupt if Hgb > 11 g/dL**
Boxed warning	Hgb targets **> 11 g/dL** → ↑ **death, MI, stroke, VTE, vascular access thrombosis** • **Tumor progression** in cancer patients
Other side effects	**Hypertension** (most common), seizures, pure red cell aplasia (rare)
Monitoring	**Hgb at least monthly** during initiation • **BP** • **Iron studies** (ESAs use up iron quickly)
Dose changes	**Do not increase more often than every 4 weeks** • If **Hgb rises > 1 g/dL in 2 weeks → reduce dose by ~25%**

ESA products

Drug	Frequency
Epoetin alfa (Epogen, Procrit, Retacrit)	**1–3 times weekly** (IV or SC); start ~**50–100 units/kg 3 times weekly**
Darbepoetin alfa (Aranesp)	**Weekly or every 2 weeks** (longer half-life); start ~**0.45 mcg/kg weekly**
Methoxy polyethylene glycol-epoetin beta (Mircera)	**Every 2 weeks or monthly**

Route: IV in hemodialysis (access available) • **SC** in non-dialysis and PD (SC is more efficient).

Causes of ESA hyporesponsiveness (frequently tested)

1. **Iron deficiency** (**most common**)
2. **Infection / inflammation**

3. **Hyperparathyroidism** (marrow fibrosis)
4. **Aluminum toxicity**
5. **B12 or folate deficiency**
6. **Inadequate dialysis**
7. Blood loss, hemolysis, malnutrition, ACE inhibitors (minor)

HIF-PH inhibitors (newer oral agents)

- **Daprodustat** (Jesduvroq), **vadadustat** (Vafseo): oral, **only for dialysis patients** (approved for patients on dialysis for **≥ 3–4 months**).
- They stimulate **endogenous EPO** production.
- Boxed warning similar to ESAs (**thrombosis, MI, stroke, death**).

Blood transfusion

- **Avoid if possible**, especially in **kidney transplant candidates** (**risk of HLA sensitization**).

8. CKD-Mineral and Bone Disorder (CKD-MBD) (high-yield)

Pathophysiology (understand the chain)

1. **↓ GFR → phosphate retention** (**hyperphosphatemia**).
2. **↑ FGF-23** (tries to excrete phosphate) → **↓ 1-α-hydroxylase**.
3. **↓ Active vitamin D (calcitriol) → ↓ GI calcium absorption**.
4. **↓ Calcium** (+ high phosphate binds calcium).
5. **↑ PTH → secondary hyperparathyroidism (SHPT)**.
6. Result: **renal osteodystrophy** (bone disease, fractures, **osteitis fibrosa cystica**) + **vascular and soft tissue calcification → cardiovascular death**.

> **Simple chain: ↑ Phosphate + ↓ Vitamin D → ↓ Calcium → ↑ PTH**

Labs and monitoring

Calcium, phosphate, PTH, alkaline phosphatase, 25-hydroxyvitamin D (frequency increases with CKD stage, starting at **G3a**).

Targets (KDIGO)

Parameter	Target
Phosphate	**Lower toward the normal range** (~2.5–4.5 mg/dL)
Calcium	**Avoid hypercalcemia** (normal range)
PTH (dialysis, G5D)	**2–9 × the upper limit of normal** (~**130–600 pg/mL**)
PTH (G3a–G5, not on dialysis)	Optimal level unknown → treat **progressively rising** or persistently high PTH
25-OH vitamin D	Correct deficiency (**< 30 ng/mL**)

Treatment step 1: Dietary phosphate restriction

- **800–1,000 mg/day**.
- **Avoid processed foods, colas (phosphoric acid), and fast food** (phosphate **additives** are almost **100% absorbed**).
- Dairy, meat, and nuts are high in phosphate.

Treatment step 2: Phosphate binders

> **All phosphate binders must be taken WITH MEALS** (they bind dietary phosphate in the gut).

Binder	Key points
Calcium carbonate (Tums)	**40% elemental calcium** • **Needs stomach acid** → less effective with **PPIs/H2RAs** • Cheap • **Hypercalcemia**, **constipation**, **vascular calcification**
Calcium acetate (PhosLo)	**25% elemental calcium** • **Binds phosphate better** than carbonate, **less acid-dependent** • Hypercalcemia
Sevelamer carbonate (Renvela) / **sevelamer hydrochloride** (Renagel)	**Calcium-free, aluminum-free** • **Lowers LDL** • GI effects • **Hydrochloride** form → **metabolic acidosis** (carbonate preferred) • **Binds** fat-soluble vitamins and drugs (**ciprofloxacin**, **mycophenolate**, **levothyroxine**)
Lanthanum carbonate (Fosrenol)	**Must be CHEWED thoroughly** (do not swallow whole: risk of GI obstruction) • Calcium-free • Nausea, vomiting
Ferric citrate (Auryxia)	**Also treats iron deficiency** • **Dark stools**, diarrhea • Iron overload
Sucroferric oxyhydroxide (Velphoro)	**Chewable** • Iron-based (minimal iron absorption) • Dark stools
Aluminum hydroxide	**Short-term only (≤ 4 weeks)** for severe hyperphosphatemia • **Aluminum toxicity**: **encephalopathy (dementia)**, **osteomalacia**, **microcytic anemia** • **Never with citrate**
Magnesium-based	Limited use (hypermagnesemia, diarrhea)
Tenapanor (Xphozah)	**Not a binder**: **NHE3 inhibitor** (blocks intestinal phosphate absorption) • **Add-on** in dialysis • **Diarrhea**

Calcium-based binder limits

- **Elemental calcium from binders ≤ 1,500 mg/day**; **total calcium intake ≤ 2,000 mg/day**.
- **KDIGO suggests restricting calcium-based binders** (risk of vascular calcification).
- **Avoid calcium binders** if **hypercalcemia**, **vascular calcification**, or **low PTH (adynamic bone disease)**.

Treatment step 3: Vitamin D

Type	Drugs	Use
Nutritional vitamin D	**Ergocalciferol (D2)**, **cholecalciferol (D3)**	**Correct 25-OH vitamin D deficiency** (< 30 ng/mL) in CKD G3–G4 (needs kidney activation)
Extended-release calcifediol (Rayaldee)	25-OH vitamin D3	SHPT in **CKD G3–G4** with vitamin D insufficiency (**not for dialysis**)
Active vitamin D / vitamin D receptor activators (VDRAs)	**Calcitriol** (Rocaltrol), **paricalcitol** (Zemplar), **doxercalciferol** (Hectorol)	**Severe and progressive SHPT** in **G4–G5**, and in **dialysis** • Do **not** need kidney activation

VDRA notes

- Main side effects: **hypercalcemia** and **hyperphosphatemia** → **correct phosphate first** and monitor Ca and Phos.
- **Paricalcitol and doxercalciferol** cause **less hypercalcemia** than calcitriol.
- **Doxercalciferol** is a **prodrug** that needs **liver activation** (not kidney).
- **Hold or reduce** if **hypercalcemia**, **hyperphosphatemia**, or PTH is oversuppressed.

Treatment step 4: Calcimimetics (dialysis patients only)

Point	Details
Drugs	**Cinacalcet** (Sensipar, **oral**, daily) • **Etelcalcetide** (Parsabiv, **IV**, **3 times weekly at the end of HD**)
Mechanism	↑ **Sensitivity of the calcium-sensing receptor** on the parathyroid → ↓ **PTH**, ↓ calcium, ↓ phosphate
Indication	**SHPT in dialysis patients only** (**not** in non-dialysis CKD: hypocalcemia risk)
Main side effect	**Hypocalcemia** (**do not start if Ca is below normal**) • Nausea, vomiting • **QT prolongation**
Cinacalcet interactions	**Strong CYP2D6 inhibitor • CYP3A4 substrate • Take with food**
Do not combine	**Cinacalcet with etelcalcetide**
Advantage	Lowers PTH **without raising calcium or phosphate** (unlike VDRAs)

Treatment step 5: Parathyroidectomy

For **severe SHPT refractory** to medical therapy.

Quick guide by lab result

Problem	Action
High phosphate	Diet + **binder with meals**
High phosphate + high calcium	**Non-calcium binder** (sevelamer, lanthanum, iron-based); **stop/reduce calcium binders and VDRAs**
High PTH + low vitamin D	**Nutritional vitamin D** first
High PTH (dialysis) + high calcium	**Calcimimetic** (rather than VDRA)
High PTH (dialysis) + low/normal calcium	**VDRA** (or calcimimetic if Ca is normal)
Low calcium	Avoid calcimimetics; consider VDRA or calcium

9. Fluid and Electrolyte Complications

Hyperkalemia

- **Causes:** ↓ excretion + **drugs** (**ACEi/ARB**, **spironolactone**, **finerenone**, **trimethoprim**, NSAIDs, heparin, beta-blockers, tacrolimus, potassium supplements, **salt substitutes (KCl)**).
- **Chronic management: low-potassium diet • loop or thiazide diuretics • correct metabolic acidosis • potassium binders** → allow continuing **RAAS inhibitors**.

Binder	Notes
Patiromer (Veltassa)	Onset ~7 h • **Separate other oral drugs by 3 hours** • **Hypomagnesemia**, constipation
Sodium zirconium cyclosilicate (Lokelma)	Onset ~1 h • **Edema** (sodium load)
Sodium polystyrene sulfonate (Kayexalate)	**Intestinal necrosis** (especially with sorbitol) • sodium load

- **Acute hyperkalemia** → calcium gluconate, insulin + dextrose, albuterol, dialysis (see AKI notes).

Edema / volume overload

- **Loop diuretics** are the **diuretics of choice** in advanced CKD (higher doses are needed as GFR falls).
- **Thiazides** are traditionally less effective when **eGFR < 30**, **but**:
- – **Metolazone** works at low GFR (often **combined with a loop** → sequential nephron blockade; watch hypokalemia).
- – **Chlorthalidone** was effective in **CKD G4** (**CLICK trial**).

Other electrolytes

Disturbance	Note
Hyperphosphatemia	See CKD-MBD
Hypocalcemia	See CKD-MBD
Hypermagnesemia	**Avoid magnesium** antacids and laxatives (e.g., milk of magnesia, magnesium citrate)
Hyponatremia	Water restriction
Fleet enema (sodium phosphate)	**Avoid**: severe **hyperphosphatemia** and **acute phosphate nephropathy**

10. Cardiovascular Disease and Hypertension

- **Cardiovascular disease is the #1 cause of death in CKD** (more likely to die of CVD than to reach dialysis).
- **Hypertension** is present in most patients and usually needs **multiple drugs**.
- – **First-line: ACEi or ARB** (especially with albuminuria).
- – **Add: diuretic** (volume), **CCB**, etc.
- **Statins** as above.
- **Aspirin** for **secondary prevention**.
- **SGLT2i**, **finerenone**, **GLP-1 RA** also reduce CV events.

11. Uremic Complications

Complication	Treatment
Uremic pruritus (itching)	**Optimize dialysis** and **phosphate control** • Emollients • **Gabapentin/pregabalin** (low dose) • Antihistamines • **UVB phototherapy** • **Difelikefalin** (Korsuva): **IV kappa-opioid agonist** for **pruritus in hemodialysis** patients
Uremic bleeding (platelet dysfunction)	**Desmopressin (DDAVP)** (fast, short-acting) • **Conjugated estrogens** (longer-acting) • **Cryoprecipitate** • Correct anemia • **Dialysis**
Uremic pericarditis	**Dialysis** (indication)
Uremic encephalopathy	**Dialysis**
Peripheral neuropathy, restless legs	
Nausea, vomiting, anorexia, metallic taste	

12. Kidney Replacement Therapy (Dialysis and Transplant)

When to start dialysis

- Based on **symptoms and complications**, not on a number alone (usually **eGFR ~5–10**).
- **AEIOU: A**cidosis, **E**lectrolytes (hyperkalemia), **I**ntoxication, **O**verload (fluid), **U**remia (pericarditis, encephalopathy, bleeding).
- Also: **malnutrition**, refractory **pruritus**, cognitive decline.

Types

Type	Key points
Hemodialysis (HD)	Usually **3 times weekly**, 3–4 h • **Access: AV fistula** is **preferred** (fewest infections and clots) > AV graft > **central venous catheter** (highest infection risk) • **Heparin** used during sessions • Complications: **intradialytic hypotension** (midodrine; hold antihypertensives before dialysis), cramps, access infection/thrombosis
Peritoneal dialysis (PD)	At home, daily, using the peritoneum • Complication: **peritonitis** (cloudy effluent, abdominal pain) → **intraperitoneal antibiotics** (preferred route) • Dextrose in dialysate → **hyperglycemia, weight gain** • **Icodextrin** → **falsely high glucose readings** with **GDH-PQQ glucometers** (risk of giving unnecessary insulin → hypoglycemia)
Kidney transplant	**Best option** (survival and quality of life) • Lifelong immunosuppression: **tacrolimus**, **mycophenolate**, **corticosteroids** (± induction)

> **"Save the vein":** In CKD G4–G5, **avoid IV lines, venipuncture, and BP measurement in the non-dominant arm** to preserve it for a future **AV fistula**. **Avoid PICC lines** if possible.

Vaccines

Vaccine	Note
Hepatitis B	**Higher dose** in dialysis patients (e.g., **Engerix-B 40 mcg** at 0, 1, 2, 6 months; **Recombivax HB dialysis formulation 40 mcg** at 0, 1, 6 months) • **Check anti-HBs** after the series and **annually** • Booster if anti-HBs < 10 mIU/mL
Influenza	Annually (**inactivated**)
Pneumococcal	Recommended
Live vaccines	**Avoid after transplant** (immunosuppression)

13. Drug Dosing in CKD

General rules

Rule	Details
Kidney function estimate	**Cockcroft-Gault** for most drug labels • **eGFR** is acceptable if **de-indexed** from BSA (mL/min, not mL/min/1.73 m²)
Loading dose	Usually **unchanged**
Maintenance dose	**Reduce the dose** and/or **extend the interval**
Dialysis	Drugs removed by dialysis (**low MW, low protein binding, small Vd, water-soluble**) → **give after the HD session**
TDM	Vancomycin, aminoglycosides, digoxin, lithium

Drugs to avoid or adjust in CKD (high-yield)

Drug	Issue in CKD
NSAIDs	**Avoid**: worsen kidney function, hyperkalemia, fluid retention
Metformin	**Contraindicated if eGFR < 30**; do not start at 30–44
Glyburide	**Avoid** (prolonged hypoglycemia) → prefer **glipizide**
Insulin	**Reduce dose** (less renal clearance)
Meperidine	**Avoid** (normeperidine → **seizures**)
Morphine, codeine	**Avoid** (active metabolites accumulate) → prefer **hydromorphone** (caution), **fentanyl**, **buprenorphine**
Tramadol	Reduce dose (max 200 mg/day if CrCl < 30); seizure risk
Gabapentin, pregabalin	**Reduce dose** (sedation, dizziness, falls)
Baclofen	**Avoid** in advanced CKD (**neurotoxicity**, encephalopathy)
Enoxaparin	**CrCl < 30 → 1 mg/kg once daily**
Fondaparinux	**Contraindicated if CrCl < 30**
Dabigatran	Mainly renal clearance → avoid in severe CKD
Rivaroxaban, edoxaban, apixaban	Adjust by CrCl (apixaban has the least renal elimination)
Warfarin	No renal adjustment, but **higher bleeding risk**
Digoxin	**Reduce dose** (toxicity: nausea, visual changes, arrhythmias; hyperkalemia worsens risk)
Spironolactone, eplerenone	**Hyperkalemia**
Nitrofurantoin	**Avoid if CrCl < 30** (ineffective, neuropathy)
TMP-SMX	**Hyperkalemia**, false creatinine rise; adjust dose
Cefepime	Accumulation → **neurotoxicity** (encephalopathy, seizures)
Imipenem	**Seizures** in renal impairment
Penicillins, most cephalosporins, fluoroquinolones	Adjust dose
Acyclovir, valacyclovir	**Neurotoxicity** (confusion, hallucinations) → adjust dose
Famotidine (H2RAs)	**CNS effects** (confusion) → adjust dose
Colchicine	Toxicity (myopathy, neuropathy) → reduce dose; avoid in dialysis
Allopurinol	**Start low** (e.g., **50–100 mg/day**), titrate slowly (hypersensitivity risk)
Bisphosphonates	**Avoid if CrCl < 30–35**
Lithium	Nephrotoxic; reduce dose or avoid
Magnesium- and phosphate-containing products	**Avoid** (milk of magnesia, **Fleet enema**)
Aluminum-containing products	**Avoid long-term** (aluminum toxicity)
Gadolinium contrast (MRI)	**Nephrogenic systemic fibrosis** if **eGFR < 30** or dialysis → avoid if possible, use **group II agents**, dialysis soon after if needed
Iodinated contrast	Contrast-associated AKI → hydration, lowest dose
Sodium phosphate bowel preps	**Acute phosphate nephropathy** → avoid

14. Monitoring Summary

Parameter	Why
eGFR and ACR	Staging and progression (at least yearly; more often in advanced stages)
Potassium and SCr	**2–4 weeks** after starting/increasing ACEi, ARB, finerenone
BP	Target SBP < 120 (KDIGO)

Parameter	Why
Hgb	Anemia (more often as CKD advances; **monthly** on ESA initiation)
TSAT, ferritin	Before and during ESA therapy
Ca, Phos, PTH, ALP, 25-OH vitamin D	CKD-MBD (from **G3a**)
Bicarbonate	Metabolic acidosis
A1c, lipids	Diabetes and CV risk
Anti-HBs	Hepatitis B immunity in dialysis patients

Exam Traps and Frequently Tested Points

1. **CKD = abnormality for > 3 months** (GFR < 60 **or** markers of damage such as ACR ≥ 30).
2. **Staging uses GFR (G1–G5) and albuminuria (A1–A3).**
3. **Diabetes is the #1 cause**, hypertension #2.
4. **Cardiovascular disease is the #1 cause of death.**
5. **ACEi/ARB** for CKD **with albuminuria**; **creatinine rise up to 30% is acceptable**.
6. **Never combine ACEi + ARB** (or with aliskiren).
7. **SGLT2i** for CKD with **eGFR ≥ 20**, **with or without diabetes**; **continue until dialysis**; initial eGFR dip is expected.
8. **Finerenone:** T2D + albuminuria, **eGFR ≥ 25**, **K ≤ 5.0**; watch **hyperkalemia**.
9. **KDIGO BP target: SBP < 120.**
10. **Metformin:** contraindicated **eGFR < 30**; do not start at **30–44**.
11. **Statin** for CKD **≥ 50 years**; **do not start in dialysis** (but continue).
12. **Protein: 0.8 g/kg/day** (non-dialysis), **1.0–1.2 g/kg/day** (dialysis).
13. **Anemia of CKD** = **↓ EPO**; **correct iron first**.
14. **Iron if TSAT ≤ 30% and ferritin ≤ 500.**
15. **Hemodialysis → IV iron**; non-dialysis → oral iron first.
16. **Iron dextran → test dose** (anaphylaxis); **ferumoxytol** → MRI interference; **ferric carboxymaltose** → hypophosphatemia.
17. **Start ESA when Hgb < 10**; **do not exceed 11.5** (reduce if > 11 per FDA).
18. **ESA boxed warning:** stroke, MI, VTE, death, tumor progression; **hypertension** is the most common side effect.
19. **#1 cause of ESA hyporesponsiveness = iron deficiency.**
20. **Do not increase ESA dose more often than every 4 weeks.**
21. **CKD-MBD chain:** ↑ Phos + ↓ vitamin D → ↓ Ca → **↑ PTH**.
22. **All phosphate binders are taken WITH MEALS.**
23. **Calcium carbonate** needs acid (less effective with PPIs); **calcium acetate** is the better binder.
24. **Sevelamer** lowers LDL; **hydrochloride** form causes **acidosis**.
25. **Lanthanum must be chewed.**
26. **Ferric citrate** also treats **iron deficiency**.
27. **Aluminum binders → max 4 weeks**; **never with citrate**.
28. **Elemental calcium from binders ≤ 1,500 mg/day.**
29. **Hypercalcemia** → switch to **non-calcium binder**, reduce VDRA.
30. **Doxercalciferol** needs **liver** activation; **paricalcitol** causes less hypercalcemia than calcitriol.
31. **Cinacalcet/etelcalcetide → dialysis only**; main side effect **hypocalcemia**; **do not combine**; cinacalcet is a **CYP2D6 inhibitor**.
32. **PTH target in dialysis: 2–9 × ULN.**
33. **Treat metabolic acidosis** if bicarbonate **< 18** (some references < 22).
34. **Patiromer** → separate other drugs by **3 hours**.
35. **Loop diuretics** are preferred for edema in advanced CKD; **metolazone** works at low GFR.
36. **Uremic bleeding → desmopressin.**
37. **HD pruritus → difelikefalin** (IV kappa-opioid agonist).

38. **AV fistula** is the preferred HD access; **"save the vein"** in the non-dominant arm.
39. **PD peritonitis → intraperitoneal antibiotics**; **icodextrin → falsely high glucose** readings.
40. **Hepatitis B vaccine → higher dose (40 mcg)** in dialysis patients.
41. **Gadolinium + eGFR < 30 → nephrogenic systemic fibrosis.**
42. **Avoid:** NSAIDs, meperidine, morphine, codeine, glyburide, Fleet enema, magnesium products, nitrofurantoin (< 30), fondaparinux (< 30).
43. **Avoid transfusions** in transplant candidates (**HLA sensitization**).

> **Note:** Some cut-off values differ slightly between references and guideline versions, such as bicarbonate thresholds, BP targets, ESA Hgb limits, and PTH ranges. **Use the numbers in the reference you are studying for the exam.**

Good luck on your exam!